R13

Code No: 117CF

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
DESIGN PATTERNS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Describe the meaning of "Intent" in design pattern.	[2]
b)	How to use a design pattern? Write the guidelines.	[3]
c)	What is the rationale behind Prototype design pattern?	[2]
d)	Explain about recursive composition.	[3]
e)	Sketch the structure of composite pattern.	[2]
f)	What are the two variations of the Adapter pattern?	[3]
g	What are the drawbacks of Mediator pattern?	[2]
h)	Write about the known uses of Iterator pattern.	[3]
i)	What can we expect from a design pattern? Explain.	[2]
j)	What is the purpose of Visitor pattern? Discuss.	[3]

PART – B

(50 Marks)

2.a) What are the patterns that are included in design pattern catalog? Write their intents.
b) What are the common causes of redesign? Explain. [5+5]

OR

3.a) How design pattern solve design problems? Explain with examples.
b) Discuss the relationship between "consequence" and "forces" in a pattern. [5+5]

4.a) The Singleton pattern uses a special method to instantiate objects. What is special about this method?
b) What are the implementation issues of prototype design pattern? Discuss. [5+5]

OR

5.a) Can we use an abstract factory for supporting multiple window system in Lexi's design? Explain.
b) What are consequences of Builder pattern? Explain Builder pattern implementation.[5+5]

6.a) Write the sample code of Flyweight pattern and explain.
b) Discuss about the participants of Adapter pattern and explain the functions of each.[5+5]

OR

7.a) What are the implementation issues of Decorator pattern? Explain.
b) Write about motivation and consequences of Façade pattern. [5+5]

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8.a) Discuss about the structure and participants of Interpreter pattern.
b) Write about the applicability and structure of Command pattern. [5+5]

OR

9.a) What are the implementation issues to be considered in Chain of Responsibility pattern? Explain.
b) Describe the implementation issues of Memento pattern. [5+5]

10.a) Write about the benefits and liabilities of the Visitor pattern.
b) What are the implementation issues of State Behavioral pattern? [5+5]

OR

11.a) Discuss the implementation issues of Strategy behavioral pattern.
b) What are the difference between the Strategy pattern and the Template Method pattern? Explain. [5+5]

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